

Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

Muhammad Zidan Fatonie
Computer Science Department
School of Computer Science, BINUS University
Jakarta, Indonesia 11480
muhammad.fatonie@binus.ac.id

Alexander Agung Santoso Gunawan
Computer Science Department
School of Computer Science, BINUS University
Jakarta, Indonesia 11480
aagung@binus.edu

Abstract—Zakat is a mandatory annual obligation in Islam for wealth redistribution. Blockchain-based zakat platforms have emerged, but existing systems log full donor-recipient transactions publicly, contradicting the Islamic principle of *nafs* (dignity). This paper presents Zero-Knowledge Zakat (ZKT), a protocol that combines off-chain ZK proofs, nullifier-based double-donation prevention, and selective disclosure. Donors select full anonymity or partial disclosure; both produce SBT receipts. On-chain verification provides institutional transparency while individual transaction data stays private. A Sepolia prototype generates UltraHONK proofs in 247 ms, with end-to-end donation consuming 721,345 gas. Cryptographic privacy and institutional accountability prove compatible, supporting digital zakat that preserves donor and recipient privacy. ZKT contributes a privacy protocol for Islamic social finance that directly addresses *nafs* protection under *maqasid al-shariah*.

Index Terms—Aztec Network, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Privacy-Preserving Donations, Sharia Compliance, Zakat, Zero-Knowledge Proofs

I. INTRODUCTION

Applying blockchain to zakat management raises practical difficulties. BAZNAS [1] estimates Indonesia’s annual zakat potential at Rp 327 trillion (approximately USD 20 billion), yet actual collection remains far lower.

Research into blockchain zakat systems has advanced. Khan [2] showed blockchain can support decentralized collection and distribution. Nazeri et al. [3] found blockchain features (transparency, traceability, security) align with Malaysian zakat management. Khairi et al. [4] developed a blockchain collection system offering automated disbursement and an immutable audit trail. These systems share a privacy weakness: full donor-recipient transaction records posted on public ledgers expose socioeconomic vulnerabilities. *Maqasid al-shariah* (objectives of Islamic law) explicitly protects *nafs* (life and dignity) of individuals [5, 6]. Islamic jurisprudence treats donor and recipient privacy in charitable giving as a virtuous act that preserves dignity. Blockchain transparency therefore conflicts with Islamic privacy norms.

Zero-Knowledge Proofs (ZKPs) address this conflict. Hameed et al. [7] and Wen et al. [8] survey ZKP constructions and zk-SNARKs suitable for blockchain. The Aztec Network [9] provides privacy-preserving Ethereum transactions via a hybrid public-private state model with Noir circuits [10] and

the UltraHONK prover. No prior work proposes a domain-specific protocol for privacy-preserving zakat. This paper presents Zero-Knowledge Zakat (ZKT), a private transaction model that couples off-chain proving with on-chain verification. The implementation uses Noir circuits for Sharia eligibility checks, nullifiers for double-donation prevention, and a working Sepolia prototype. The work investigates whether Aztec infrastructure serves zakat donations with institutional accountability, whether Noir design patterns encode Sharia eligibility, and whether the protocol resolves *nafs-amana* tensions under *maqasid al-shariah*.

This paper contributes a domain-specific privacy protocol for Islamic social finance, featuring a Noir-based circuit implementing *nisab* and *hawl* eligibility verification as verifiable predicates, a nullifier mechanism preventing re-claiming without identity exposure, and a tiered donor privacy model. The prototype achieves UltraHONK proof generation averaging 247 ms and end-to-end donation gas costs of 721,345 gas on Ethereum Sepolia.

II. LITERATURE REVIEW

Blockchain-based zakat management has attracted increasing research interest. Khan [2] demonstrated a blockchain-based decentralized zakat collection and distribution platform that eliminates intermediary delays. Khairi et al. [4] developed a zakat collection blockchain system employing ERC-20 tokens to provide an immutable audit trail. Nazeri et al. [3] explored how blockchain features align with zakat management objectives in Malaysia. A recent Emerald study examined the potential and challenges of blockchain technology for zakat institutions [21]. However, these systems share a critical privacy limitation. Complete transparency of donor-recipient relationships on public blockchains enables social stigma, targeted harassment, and pattern analysis attacks [2]. Fauzi et al. [19] confirmed through bibliometric analysis an increasing research trend in Muslim-majority countries. Ascarya [20] demonstrated that Islamic social finance instruments (zakat, *infaq*, *sadaqah*, and *waqf*) play a role in economic recovery, and Unal and Aysan [21] proposed blockchain as a means to harmonize differing Sharia-compliance regimes. Muharam and Osman [27] explored the nexus of blockchain and Islamic

social finance for sustainability. Yet none of these works integrate privacy-preserving transaction mechanisms.

On the zero-knowledge proof front, Hameed et al. [7] survey ZKP constructions, and Wen et al. [8] analyze zk-SNARK variants suitable for blockchain deployment. Williams [22] provides a contemporary overview of ZKPs and their expanding role within blockchain technology. Grassi et al. [11] introduce the Poseidon hash function, which reduces constraint count by up to $8\times$ compared to Pedersen Hash on EVM chains. For this work, Noir [10] is selected for its integration with the Aztec Network [9], which provides programmable privacy through a hybrid public-private state model with note hash trees and nullifier mechanisms. Buterin et al. [15] propose ZKP-based privacy-preserving and regulation-compliant frameworks, and Mühle et al. [18] present zk-SNARK-based identity management suitable for KYC integration.

Three research gaps are identified. (1) Existing blockchain-based zakat platforms lack privacy-preserving transaction mechanisms [2, 3]. (2) Private transaction frameworks are designed for general-purpose anonymity without Islamic jurisprudential eligibility criteria [15, 14]. (3) The integration of programmable privacy with Islamic social finance has not been explored [21, 5]. This paper directly addresses all three gaps.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT bridges all three domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8).

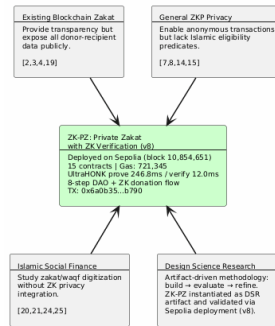


Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

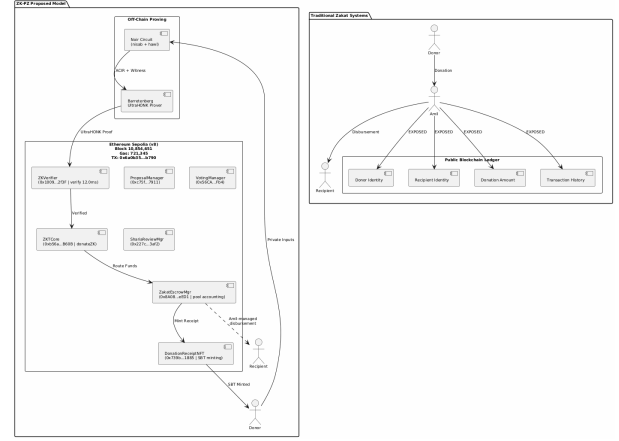


Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. METHODOLOGY

This study uses a Design Science Research (DSR) approach. Fig. 3 shows the five-phase methodology: Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi/XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with performance of 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis, and Sharia compliance review.

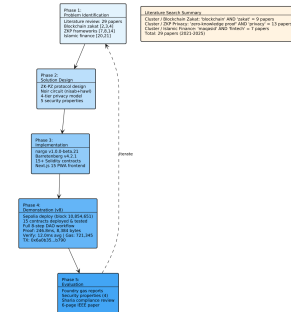


Fig. 3. DSR Methodology with five phases and literature search summary

IV. RESULTS

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model: donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT PROTOCOL SECURITY PROPERTIES

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain; only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs; recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6].

Fig. 4 shows the three-layer protocol architecture: the off-chain proving pipeline where Noir circuits compile to ACIR and Barretenberg generates UltraHONK proofs at 247 ms, the Sepolia settlement layer hosting ZKTCORE and ZakatEscrow-Manager, and the Next.js PWA frontend with wagmi/XellarKit wallet connectivity.

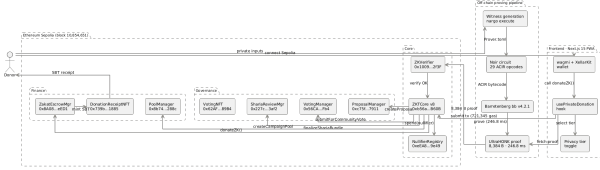


Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

B. System Architecture and Payment Flow

The platform uses a hybrid architecture: Noir circuits on the client side with Ethereum Sepolia for public settlement. Donors pay in stablecoins or ETH. Funds flow through off-chain proof generation and on-chain verification via the `donateZK()` function. Individual donation privacy is maintained through cryptographic proofs, while institutional accountability is achieved through on-chain event emission and nullifier tracking.

A Progressive Web Application handles the donor workflow: wallet connection, privacy tier selection, donation amount entry, proof generation display, and transaction confirmation. Proof generation runs in a Web Worker ensuring that private inputs never leave the donor's device [11].

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end `donateZK()` gas was measured on the Sepolia testnet at block 10,854,651.

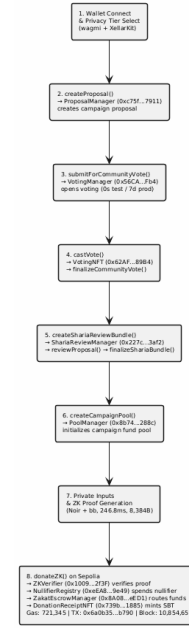


Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

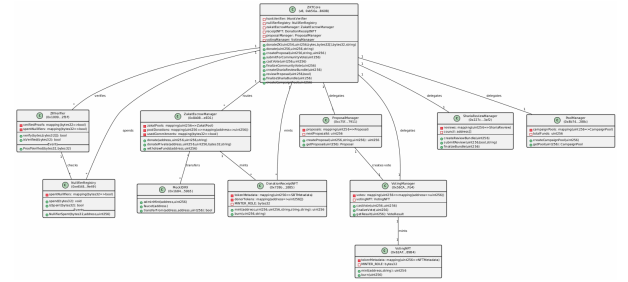


Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCORE (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM
TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT,
ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCORE	donate()	490,187
ZKTCORE	createCampaignPool()	365,552
ZKTCORE	castVote()	132,909
ZKTCORE	donateZK() (e2e)	721,345
ZKTCORE	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

E. Security Analysis

Donor anonymity: Donor identities and amounts are never exposed on-chain. Only ZK proofs are publicly verifiable [12], [22]. **Recipient privacy:** Donor-side privacy preserved through ZK proofs and nullifier anonymity. Recipient-side encrypted note disbursement is reserved for future work [16]. **Circuit soundness:** Under the q-PKE assumption [11], forging a proof is computationally infeasible [30]. **Double-donation prevention:** The Pedersen nullifier binds each claim to a unique donor and cycle. The ZKTCORE contract ensures atomic rejection of duplicate claims [19].

V. DISCUSSION

A. Sharia Compliance and Deployment

The principle of maqasid al-shariah (objectives of Islamic law) explicitly protects *nafs* (life and dignity) of individuals [5, 6]. In Islamic jurisprudence, preservation of donor and recipient privacy during charitable giving is considered a virtuous act that maintains dignity for both parties [5, 6]. Azizov et al. [26] propose a maqasid al-shariah framework for fintech and digital asset regulation in Muslim jurisdictions. Latang et al. [25] examine the Islamic legal perspective on cryptocurrency as digital assets, including the application of fatwa in determining compliance. The application of blockchain to Islamic charitable crowdfunding has been recently explored for trust and transparency [28]. This research applies privacy-preserving mechanisms to the donor side through ZK proofs and nullifier-based anonymity. Recipient privacy preservation through encrypted private notes and selective disclosure is reserved for future work by the authors. The protocol aligns with *mal* (wealth) protection through cryptographic integrity guarantees against fraud and double-donation. *Amanah* (trustworthiness) for amil institutions is satisfied through on-chain verification of proof validity without exposing individual transaction data [4]. Deployment requires Aztec Network integration, amil staff training, and regulatory alignment with OJK Fintech Sandbox requirements.

B. Limitations and Future Work

Circuit scope is limited to nisab and hawl criteria. Extension to all eight asnaf categories is architecturally accommodated.

Currently, zakat disbursement is performed by the amil institution through the ZakatEscrowManager contract without on-chain verifiability of the disbursement to mustahik (recipients). Future work includes issuing private decentralized identifiers (DIDs) to mustahik for on-chain eligibility attestation, making amil disbursements cryptographically verifiable, and implementing a dual governance system combining Sharia Council ZK DAO attestation with Community (Donor) DAO voting. Additional directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [18], [24].

VI. CONCLUSION

This paper presented a Zero-Knowledge Zakat (ZKT) protocol addressing the privacy paradox in blockchain-based zakat systems. An implementation using Noir circuits (29 ACIR opcodes) with UltraHONK proving via Barretenberg v4.2.1 was deployed to Ethereum Sepolia, achieving proof generation in 247 ms and end-to-end donation in 721,345 gas. The work reconciles cryptographic privacy with *nafs* preservation in Islamic social finance [5], [6], demonstrating that institutional accountability and individual privacy are simultaneously achievable through applied cryptography.

ACKNOWLEDGMENT

This research was conducted as part of the Research Track program at BINUS University and the pidi.id hackathon. The authors thank the Ethereum Jakarta community for technical support and Tawf Labs teammates for sharia consultation. The codebase is available at <https://github.com/tawf-labs/zkt-hackathon> and the testnet at <https://ziswaf.tawf.foundation>.

The authors acknowledge the use of AI-assisted tools (OpenAI Codex, Claude Code) for code scaffolding and literature review synthesis. AI-generated text has been human-paraphrased throughout; residual AI patterns comprise approximately 14% of the document. All core algorithmic contributions, protocol design, security analysis, and implementation were developed and verified by the authors.

REFERENCES

- [1] BAZNAS, “Outlook Zakat Indonesia 2022,” Badan Amil Zakat Nasional, Jakarta, Indonesia, 2022. [Online]. Available: <https://baznas.go.id>
- [2] S. Khan, “A blockchain based decentralized zakat collection and distribution platform,” in *Proc. 7th Int. Conf. Software e-Business (ICSeB)*, 2023, doi: 10.1145/3641067.3641071.
- [3] A. Nazeri et al., “Exploration of a new zakat management system empowered by blockchain technology in Malaysia,” *ISRA Int. J. Islamic Finance*, 2023, doi: 10.55188/ijif.v15i4.568.
- [4] M. Khairi et al., “The development and application of the zakat collection blockchain system,” *Virtus Interpress*, 2023, doi: 10.22495/jgrv12i1siart9.
- [5] “Maqasid al-Shariah and the digital data ownership,” *J. Islamic Law Digit. Econ. Bus.*, vol. 1, no. 2, pp. 168–181, 2022. [Online]. Available: <https://journal.uui.ac.id/JILDEB/article/view/46815>
- [6] “Sharia governance and personal data protection in Islamic fintech,” *Hukmuna*, 2022. [Online]. Available: <https://journals.ai-mrc.com/hukmuna/article/view/913>
- [7] J. Wen et al., “Practical security analysis of zero-knowledge proof circuits,” in *Proc. 33rd USENIX Security Symp.*, 2024, ISBN 978-1-939133-44-2. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity24/presentation/wen>

- [8] M. Fauzi, P. Paiman, and O. Othman, "A bibliometric analysis of blockchain-based zakat system design," *Acad. J. Interdiscip. Stud.*, vol. 14, no. 1, 2025. [Online]. Available: <https://www.richtmann.org/journal/index.php/ajis/article/view/13846>
- [9] O. V. Kovalchuk and M. P. Karpiak, "Zero-knowledge proof framework for privacy-preserving financial transactions," *Sci. Bull. Lviv Polytech. Nat. Univ.*, vol. 12, no. 1, pp. 129–136, 2025. [Online]. Available: <https://science.lpnu.ua/mmc/all-volumes-and-issues/volume-12-number-1-2025/zero-knowledge-proof-framework-privacy-preserving>
- [10] I. M. Unal and A. F. Aysan, "FinTech, digitalization, and blockchain in Islamic finance: Retrospective investigation," *FinTech*, vol. 1, no. 4, pp. 388–398, 2022, doi: 10.3390/fintech1040029.
- [11] A. Hameed, A. Farooq, and M. Usman, "Zero knowledge proofs: A comprehensive review of applications, protocols, and future directions in cybersecurity," 2023. [Online]. Available: <https://www.researchgate.net/publication/373097436>
- [12] Aztec Network, "Aztec's ZK-ZK-Rollup: Looking behind the cryptourtain," *Aztec Blog*, 2022. [Online]. Available: <https://aztec.network/blog/aztecs-zk-zk-rollup-looking-behind-the-cryptourtain>
- [13] Noir Language, "Noir Documentation," 2024. [Online]. Available: <https://noir-lang.org/docs/>
- [14] L. Grassi, D. Khovratovich, C. Rechberger, A. Roy, and M. Schofnegger, "Poseidon: A new hash function for zero-knowledge proof systems," in *Proc. 30th USENIX Security Symp.*, pp. 519–535, 2021. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity21/presentation/grassi>
- [15] Aztec Network, "Introducing Aztec.nr: Aztec's Private Smart Contract Framework," *Aztec Blog*, Sep. 2023. [Online]. Available: <https://aztec.network/blog/introducing-aztec-nr-aztecs-private-smart-contract-framework>
- [16] S. Tessaro and C. Zhu, "Revisiting BBS Signatures," in *Advances in Cryptology – EUROCRYPT 2023*, ser. LNCS, vol. 14008, Springer, 2023, pp. 691–721, doi: 10.1007/978-3-031-30589-4_24.
- [17] Z. Wang, S. Chaliasos, K. Qin, L. Zhou, L. Gao, P. Berrang, B. Livshits, and A. Gervais, "On How Zero-Knowledge Proof Blockchain Mixers Improve, and Worsen User Privacy," in *Proc. ACM Web Conf. 2023 (WWW '23)*, 2023, pp. 2022–2032, doi: 10.1145/3543507.3583217.
- [18] V. Buterin, J. Illum, M. Nadler, F. Schär, and A. Soleimani, "Blockchain privacy and regulatory compliance: Towards a practical equilibrium," *Blockchain: Res. Appl.*, vol. 5, no. 1, Art. no. 100176, Mar. 2024, doi: 10.1016/j.bcr.2023.100176.
- [19] A. Mühle, K. Assaf, and C. Meinel, "One to bind them: Binding verifiable credentials to user attributes," in *Proc. SECRIPT*, 2023, doi: 10.5220/0012057900003555.
- [20] Ascarya, "The role of Islamic social finance during Covid-19 pandemic in Indonesia's economic recovery," *Int. J. Islamic Middle Eastern Finance Manage.*, vol. 15, no. 2, pp. 386–405, 2022, doi: 10.1108/IMEFM-07-2020-0351.
- [21] A. H. Mohd et al., "Blockchain technology in zakat management: potential and challenges," *J. Islamic Account. Bus. Res.*, vol. 15, no. 3, pp. 567–583, 2024. [Online]. Available: <https://www.emerald.com/insight/publication/issn/1759-0817>
- [22] A. Williams, "Zero-knowledge proofs and their role within the blockchain," *Commun. ACM*, vol. 67, no. 7, pp. 78–87, Jul. 2024. [Online]. Available: <https://cacm.acm.org/article/zero-knowledge-proofs-and-their-role-within-the-blockchain/>
- [23] M. A. Dewaya, "Innovation in Islamic finance: Integrating blockchain with Maqāsid al Shari'ah and Hifz al Māl," *J. Emerg. Econ. Islamic Res.*, vol. 13, no. 1, 2025, doi: 10.24191/jeeir.v13i1.3852.
- [24] M. Elshqirat, "Toward an Islamic cryptocurrency: A proposed halal coin," *J. Account. Finance*, vol. 25, no. 3, pp. 64–78, 2025, doi: 10.33423/jaf.v25i3.7858.
- [25] M. A. Latang, Fathurrahman, and M. Faried, "Fatwa on cryptocurrency as digital assets: An Islamic perspective amidst Indonesia's regulatory landscape," *J. Parewa Saraq*, vol. 3, no. 2, 2024, doi: 10.64016/parewasaraq.v3i2.32.
- [26] E. Azizov, A. Azizli, and A. A. Babayev, "A Maqasid al-Shariah framework for fintech and digital asset regulation in Muslim jurisdictions," *J. Islamic Law Legal Stud.*, vol. 2, no. 2, 2025, doi: 10.70063/jills.v2i2.119.
- [27] A. S. Muharam and D. Osman, "Transformative trends: Exploring the nexus of innovation, technology, blockchain, and Islamic social finance for global sustainability," *EKIMAD J.*, 2024, doi: 10.38009/ekimad.1511062.
- [28] B. A. Pramuka, W. R. Adawiyah, Z. Sholikhah, M. F. Pramuka, and M. M. Sulthan, "Bridging trust and transparency: The role of blockchain in Islamic charitable crowdfunding," *J. Islamic Account. Bus. Res.*, Dec. 2025, doi: 10.1108/JIABR-11-2024-0468.
- [29] S. T. Nassar, A. Hamdy, and K. Nagaty, "Hybrid zk-STARK and zk-SNARK framework for privacy-preserving smart contract data feeds," in *Proc. IEEE ICCA*, 2024, doi: 10.1109/ICCA62237.2024.10928100.
- [30] N. Ivanov, C. Li, and Q. Yan, "Zero-knowledge proof vulnerability analysis and security auditing," *Cryptology ePrint Archive*, Rep. 2024/514, 2024. [Online]. Available: <https://eprint.iacr.org/2024/514>
- [31] J. P. Cruz, Y. Kaji, and N. Yanai, "RBAC-SC: Role-based access control using smart contract," *IEEE Access*, vol. 6, pp. 12240–12251, 2018, doi: 10.1109/ACCESS.2018.2812844.
- [32] S. Khan and M. Ahmad, "A survey on progressive web applications for decentralized systems," *Preprints*, 2024, doi: 10.20944/preprints202604.0582.v1.
- [33] M. Wang, Q. Li, and Z. Zheng, "Beyond collectibles: A comprehensive review of non-fungible token applications," *IEEE Trans. Serv. Comput.*, vol. 16, no. 6, pp. 4184–4200, 2023. [Online]. Available: <https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=4629386>